

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Kovvuru Deepika	Reg. No.:	192371042
Section / Batch:	4th year -CSE	Date:	23/07/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Search Engine Optimization System

A search engine processes queries using the recurrence $T(n) = T(n/2) + 1$. Apply the substitution method to expand the recurrence step-by-step. Derive the final time complexity and simplify the result. Analyze how the recursion depth influences the performance of the system. Interpret the result using asymptotic notation and explain why this approach is efficient for large datasets.

Q2. Rank Comparison System

A university ranking system compares each student's performance with every other student to determine overall rankings using a pairwise comparison approach. Explain how nested comparisons influence the time complexity of the algorithm and describe the growth rate as the number of students increases. Determine the asymptotic time complexity using Big-O notation and justify your

reasoning. Additionally, analyze the space complexity of the system, considering storage of intermediate comparisons. Interpret how this approach impacts performance for large datasets and discuss its efficiency.

Code of Conduct

I K.Deepika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: k.deepika(192371042)

1.Search Engine optimization

Code:

```
def search_system(n, depth=0):
    print(f"Depth {depth}: Processing size n = {n}")
    if n <= 1:
        return 1
    return search_system(n // 2, depth + 1) + 1

# Example run with dataset size n = 16
n = 16
total_steps = search_system(n)
print(f"\nTotal operations executed for n={n}: {total_steps}")
```

OUTPUT:

```
Output
Depth 0: Processing size n = 16
Depth 1: Processing size n = 8
Depth 2: Processing size n = 4
Depth 3: Processing size n = 2
Depth 4: Processing size n = 1

Total operations executed for n=16: 5

=== Code Execution Successful ===
```

2. Rank Comparsion System

Code:

```
def pairwise_rank_system(students):
    n = len(students)
    comparisons_count = 0
    scores = {student: 0 for student in students}

    print(f"Starting pairwise comparisons for n = {n} students...\n")
    for i in range(n):
        for j in range(i + 1, n):
            comparisons_count += 1
            p1, p2 = students[i], students[j]
            if p1["score"] > p2["score"]:
                scores[p1["name"]] += 1
            else:
                scores[p2["name"]] += 1
            print(f"Comparison #{comparisons_count}: {p1['name']} vs {p2['name']}")

    return scores, comparisons_count

# Execution with 4 students
students = [
    {"name": "Alice", "score": 88},
    {"name": "Bob", "score": 95},
    {"name": "Charlie", "score": 72},
    {"name": "Diana", "score": 90}
]

scores, total_ops = pairwise_rank_system(students)
print(f"\nExecution Complete. Total Comparisons: {total_ops}")
print("Final Scoreboard:", scores)
```

OUTPUT:

```
=====
=====
Starting pairwise comparisons for n = 4 students...

Comparison #1: Alice (88) vs Bob (95)
Comparison #2: Alice (88) vs Charlie (72)
Comparison #3: Alice (88) vs Diana (90)
Comparison #4: Bob (95) vs Charlie (72)
Comparison #5: Bob (95) vs Diana (90)
Comparison #6: Charlie (72) vs Diana (90)

Execution Complete!
Total Pairwise Comparisons Performed: 6
Final Scoreboard (Wins): {'Alice': 1, 'Bob': 3, 'Charlie': 0, 'Diana': 2}
=====
=====
```

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____